

e-SentriCity

**Smart detection of accidents,
threats and emergencies.**

DETECT

ANALYZE

TRACK

VERIFY

RESPOND

**WOMEN-SAFETY FIRST.
ACCIDENT-READY.**

TEAM

CLASHERS

NIRMAAN 2026

EDGE AI & DEEPTech

DETECT.
ANALYZE. TRACK.
VERIFY. **RESPOND.**

THE GAP

**SECONDS LOST.
LIVES AT RISK.**

Manual reporting creates delays

Victims may be unable to report

Detection, evidence, location and
response are fragmented

We need faster detection and
verified escalation



Our Solution

A Unified Multi-Level Verification Pipeline

01

Automatic incident detection

Sensors detect anomalies in real time



02

Context capture

Cameras capture live visual evidence



03

AI analysis & tracking

Edge AI analyzes, tracks and assesses risk



04

Human verification

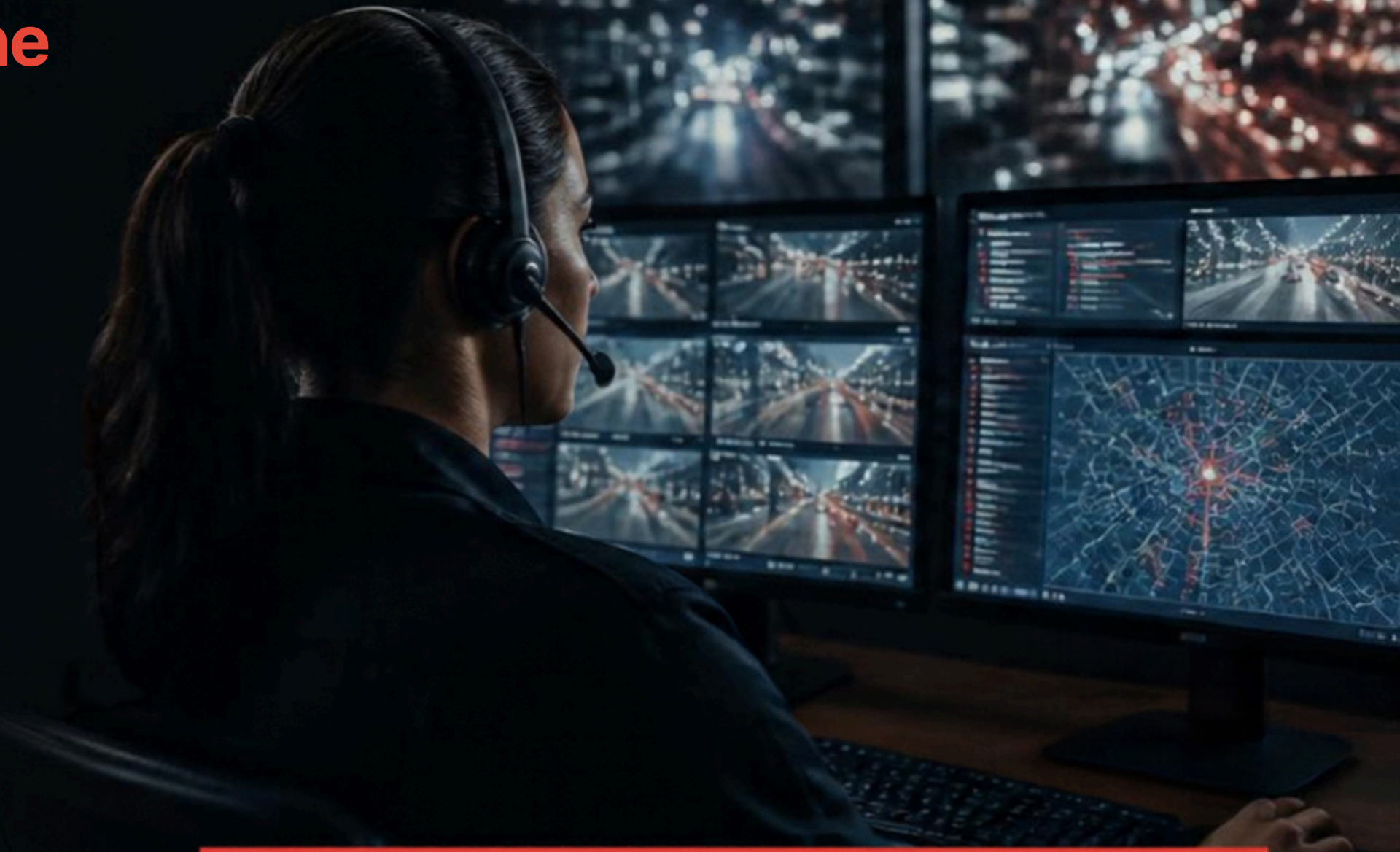
Operator validates and confirms incident



05

Emergency response

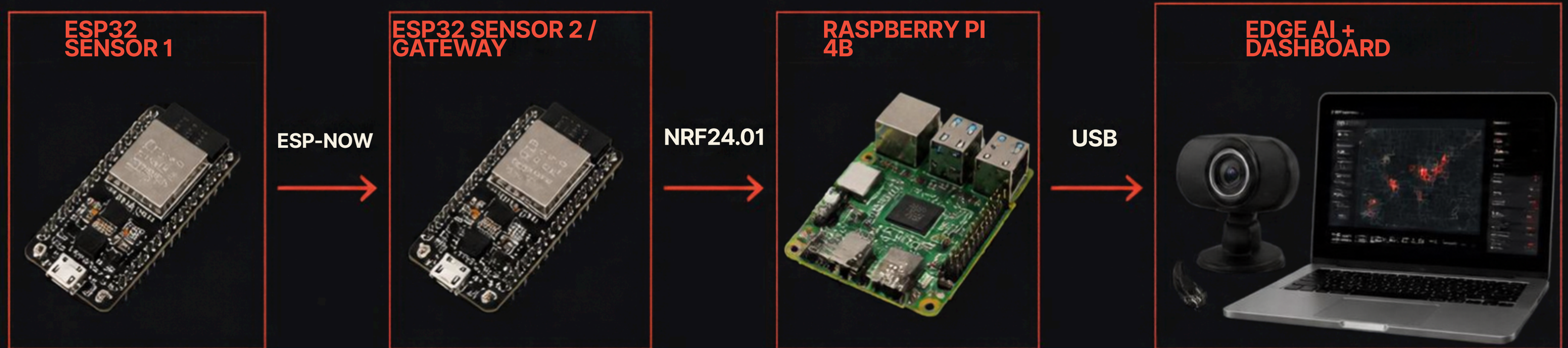
Verified incidents trigger instant response



**Multi-level
verification. Fewer
false escalations.**

HOW e-Sentricity WORKS

EDGE AI. DISTRIBUTED SENSING. REAL-TIME RESPONSE.



sensors detect
anomalies locally



data sent via
ESP-NOW



gateway forwards
over nRF24.01



edge AI processes
video + sensor data



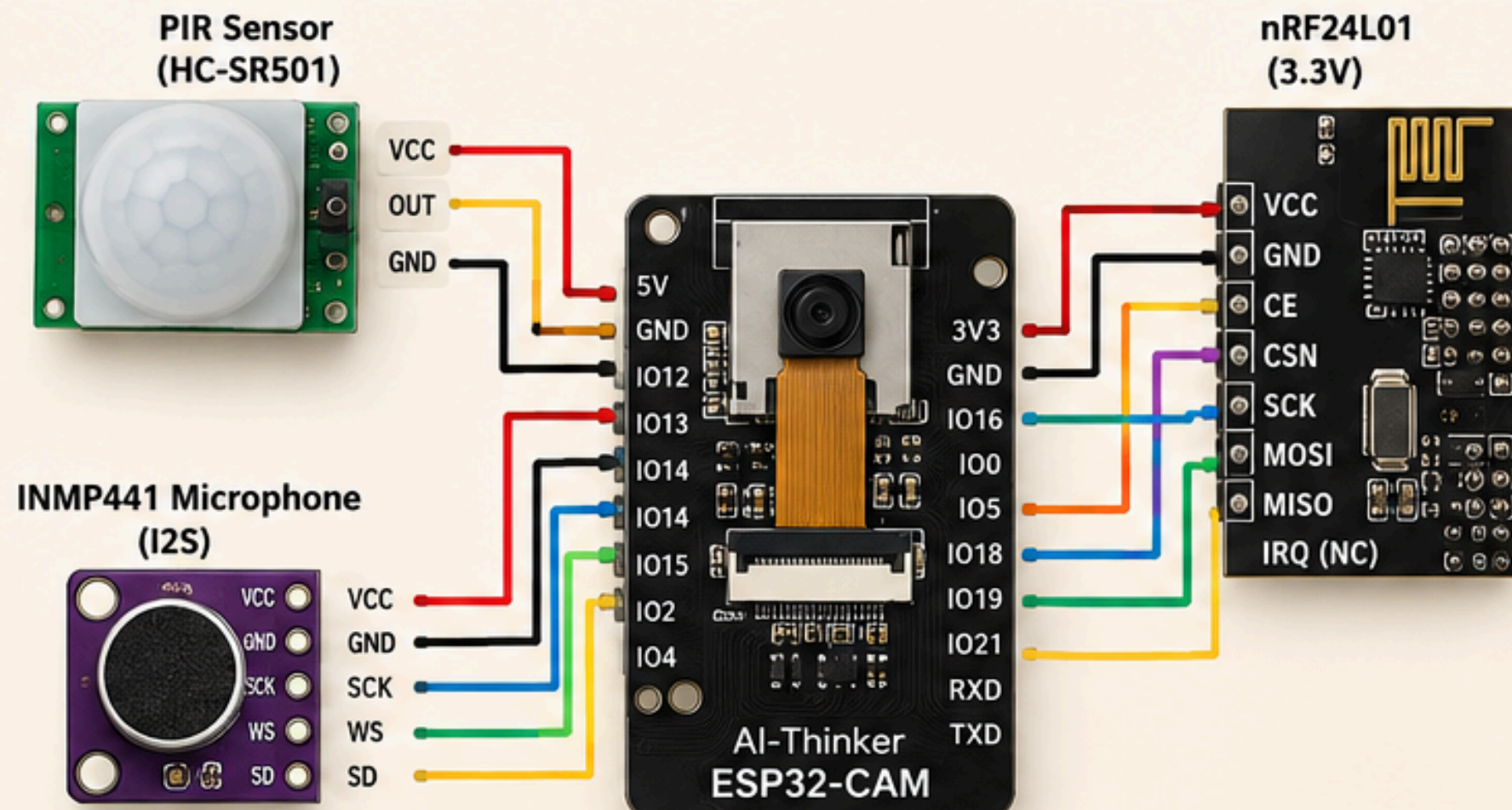
insights shown
on dashboard

Circuit Diagram

EDGE AI. DISTRIBUTED SENSING. REAL-TIME RESPONSE.

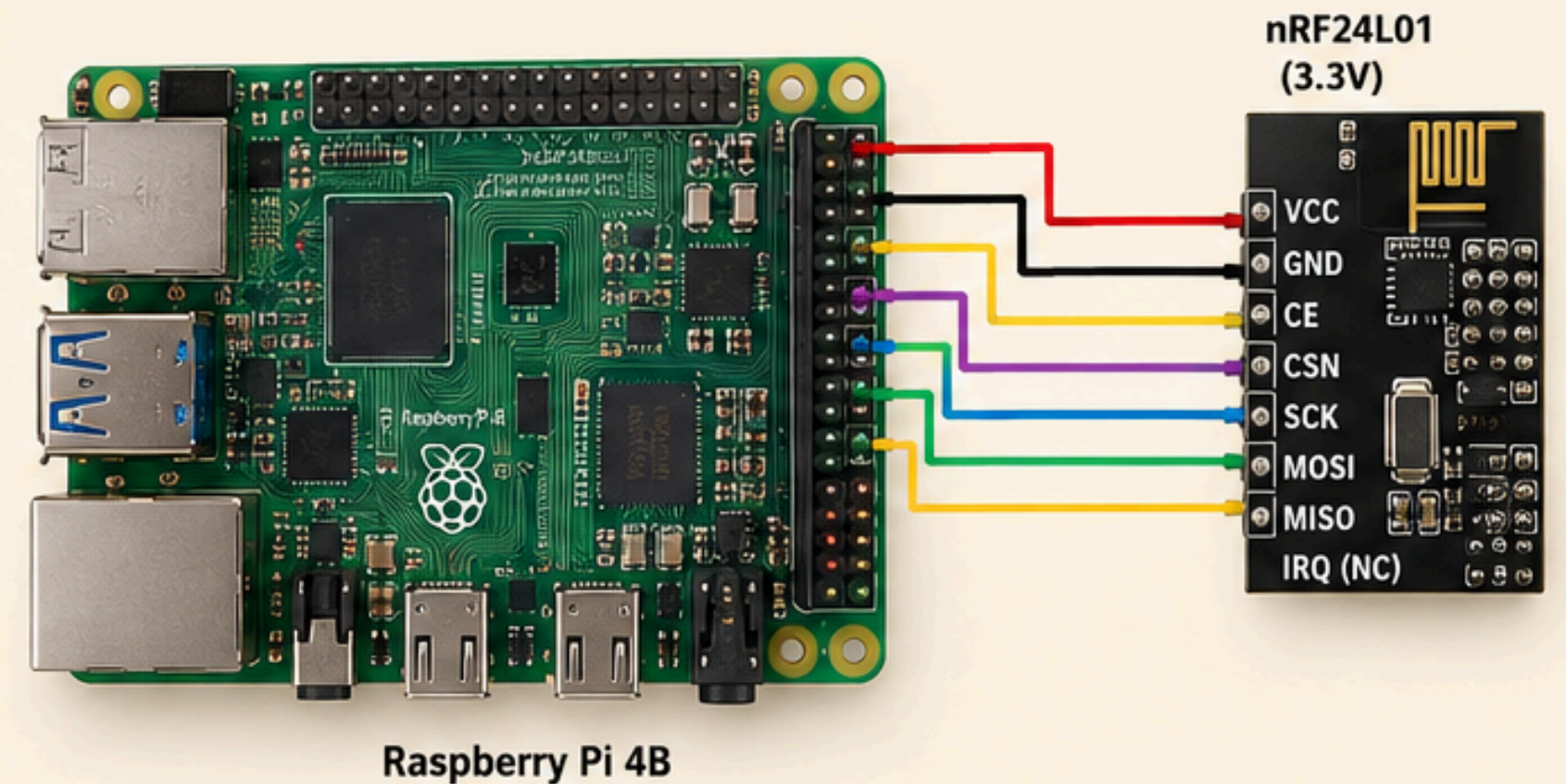
1. ESP32 NODE (AI-Thinker ESP32-CAM)

Camera + PIR + Microphone + nRF24L01



2. RASPBERRY PI NODE (Raspberry Pi 4B)

nRF24L01 + Edge AI Processing + Dashboard



TWO SCENARIOS, ONE PLATFORM.

WOMEN SAFETY FIRST

Potential distress
Suspicious forced movement
Possible abduction indicators

SHARED RESPONSE ARCHITECTURE

Location
Evidence
Tracking

ACCIDENT / EMERGENCY

Potential road accident
Medical response needed

AFTER HUMAN VERIFICATION

EMERGENCY SERVICES AND CONTROL ROOM



24-HOUR PROTOTYPE & DEMO

EDGE NODES CAPTURE.
GATEWAY ANALYSES.

AI DETECTS AND TRACKS.

DASHBOARD SHOWS LIVE
INCIDENTS & EVIDENCE.

SIMULATED RESPONSE
COMPLETES THE LOOP.



IMPACT & NEXT STEP

Faster detection and verified response

Distributed nodes scale to more communities

Smarter tracking with fewer false alerts

Field-ready hardware and real dispatch integration

e-SentriCity: Detect Real-time. Verify intelligently.
Respond immediately.